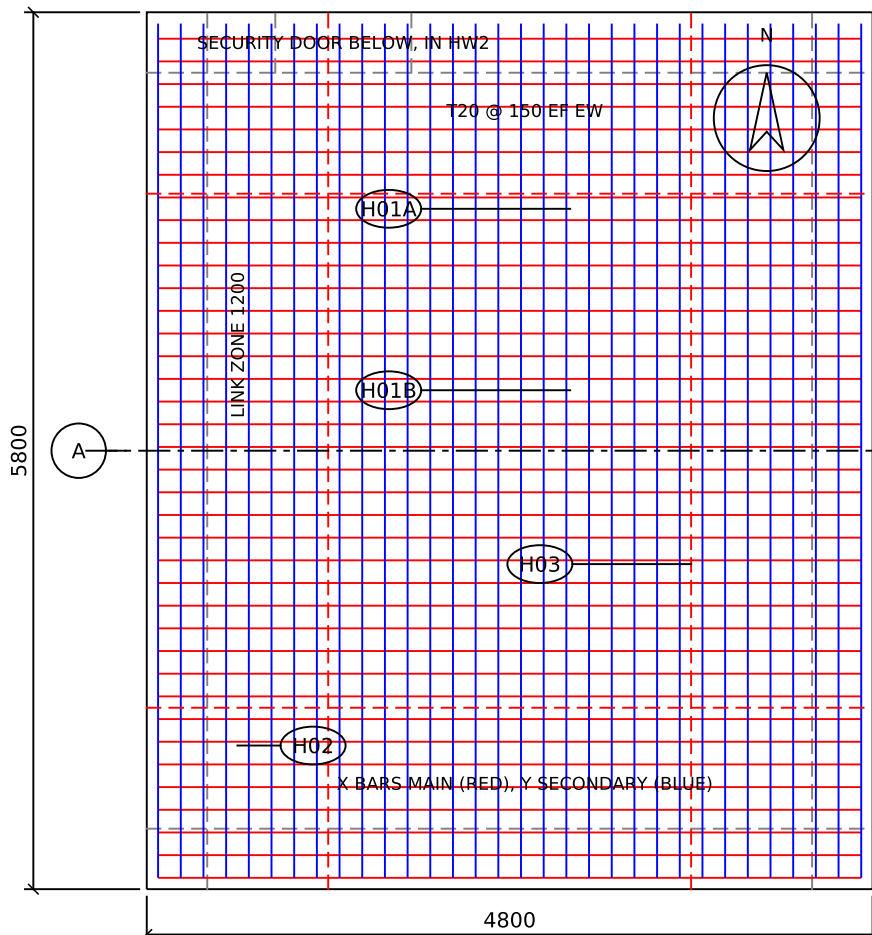
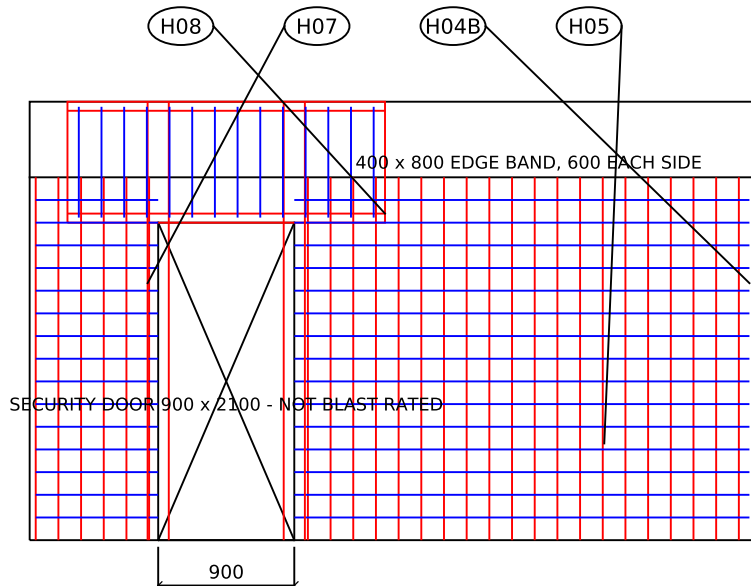
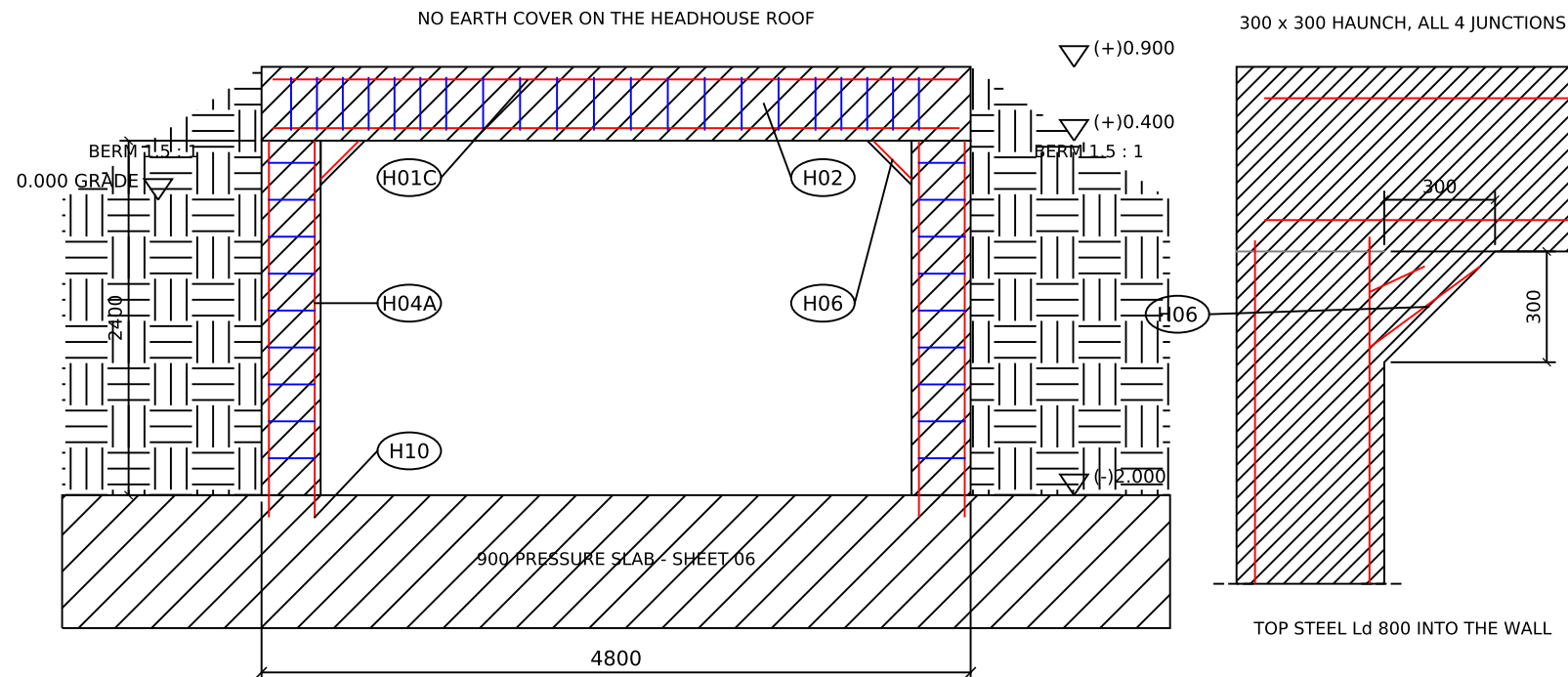




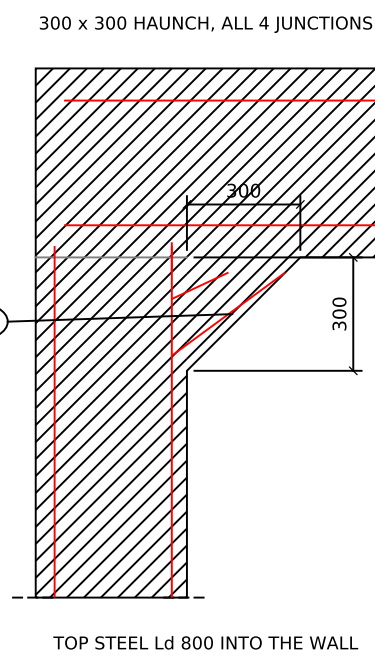
1 HEADHOUSE - ROOF REINFORCEMENT PLAN
1 : 50 ROOF 500 THK, T20 @ 150 EF EW (2094 mm2/m), COVER 75



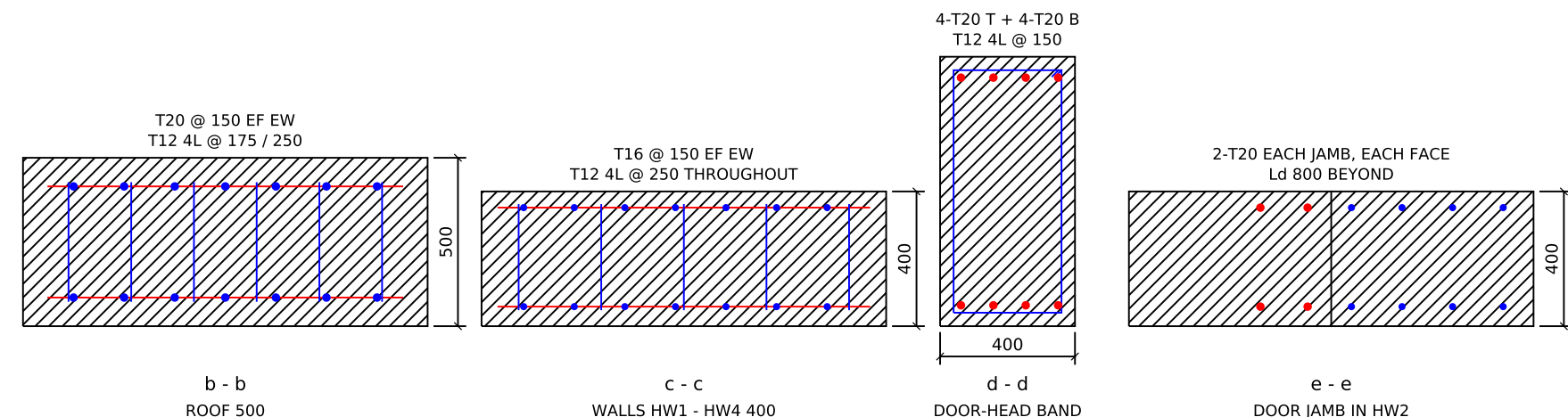
3 WALL HW2 - INTERNAL ELEVATION
1 : 50 400 THK, T16 @ 150 EF EW, AT THE SECURITY DOOR



2 HEADHOUSE - SECTION A-A
1 : 50 CUT IN X, LOOKING NORTH



5 WALL / ROOF HAUNCH
1 : 20 DIAGONAL T16 @ 150



4 SECTIONS - ROOF, WALL, DOOR HEAD AND JAMB
1 : 20 COVER 75 ROOF, 50 EARTH FACE, 40 INTERNAL

HEADHOUSE - ELEMENT SCHEDULE

ELEMENT	THK	COVER	d	MAIN REINFORCEMENT	LINKS
ROOF - END 1200 EACH SIDE	500	75	415	T20 @ 150 EF EW (2094 mm2/m)	T12 4L @ 175
ROOF - MID ZONE	500	75	415	T20 @ 150 EF EW	T12 4L @ 250
WALLS HW1 - HW4	400	50 / 40	342	T16 @ 150 EF EW (1340 mm2/m)	T12 4L @ 250 THROUGHOUT
DOOR JAMBS, HW2	-	40	-	2-T20 EACH JAMB, EACH FACE, Ld 800 BEYOND	-
DOOR-HEAD EDGE BAND	400 x 800	40	742	4-T20 TOP + 4-T20 BOTTOM	T12 4L @ 150
WALL / ROOF HAUNCH	300 x 300	-	-	DIAGONAL T16 @ 150	-
THE WALLS ARE SYMMETRICAL FOR 383 kPa EITHER FACE. ROOF SPANS 4000 CLEAR, WALLS 2400 CLEAR					

BAR MARK SCHEDULE - HEADHOUSE ROOF 500 AND WALLS 400

MARK	BAR	SPACING	LOCATION	CUT mm	No.
H01A	T20	150	BOTTOM, MAIN (SPANNING 4000 IN X)	4650	38
H01B	T20	150	BOTTOM, SECONDARY (Y)	5650	32
H01C	T20	150	TOP, MAIN (X)	4650	38
H01D	T20	150	TOP, SECONDARY (Y)	5650	32
H02	T12	175 x 175	4-LEGGED LINKS @ 175, END 1200 EACH SIDE	1240	672
H03	T12	250 x 250	4-LEGGED LINKS @ 250, MID ZONE	1240	240
H04A	T16	150	VERTICAL, EACH FACE	3825	260
H04B	T16	150	HORIZONTAL RINGS, EACH FACE	10200	68
H05	T12	250 x 250	4-LEGGED LINKS @ 250, ALL FOUR WALLS	1140	1404
H06	T16	150	HAUNCH DIAGONALS, 300 x 300	1704	130
H07	T20	-	JAMB BARS, 2-T20 EACH JAMB EACH FACE	3700	8
H08	T20	-	4-T20 TOP + 4-T20 BOTTOM	3700	8
H09	T12	150	4-LEGGED LINKS @ 150	2320	28
H10	T16	150	WALL STARTERS, EACH FACE	2335	260

HEADHOUSE - DESIGN SUMMARY

ELEMENT	BASIS	M or V	Ast,req / tau_c	PROVIDED	UTILISATION
ROOF 500 - ONE-WAY STRIP	w Ln2 / 16	396.5 kNm/m	1879 mm2/m	2094 (T20 @ 150 EF EW)	90 % *** ADOPTED ***
ROOF 500 - IS 456 TABLE 26	CASE 1, alpha x= 0.045	285.5 kNm/m	-	as above	65 %
ROOF 500 - YIELD LINE	FIXED 4 EDGES, ISOTROPIC	162.2 kNm/m	-	as above	37 %
WALLS 400	w Ln2 / 16 AT 383 kPa	137.9 kNm/m	766 ; Cl. 32.5(b) min 800	1340 (T16 @ 150 EF EW)	59 %
SHEAR - tau_c IS READ AT THE STATIC M35 VALUE. IS 4991 Cl. 10.3.1.1 FORBIDS A DYNAMIC INCREASE ON SHEAR					
ROOF SHEAR AT d	V 628.4 kN/m	tau_v 1.514	tau_c 0.502	T12 4L @ 175 / 250	LINKS MANDATORY
WALL SHEAR AT d	V 328.6 kN/m	tau_v 0.961	tau_c 0.444	T12 4L @ 250	Cl. 26.5.1.5 256 GOVERNS

CONSTRUCTION REQUIREMENTS THAT ARE DESIGN OUTPUTS, NOT OPTIONS

- THE WALLS ARE REINFORCED SYMMETRICALLY FOR 383 kPa ACTING EITHER FACE. The inner security door is NOT blast rated and the entry stairwell is expected to be lost, so the headhouse fills and the walls are pushed OUTWARDS. A stated requirement, not an accident of detailing.
- HW3 HAS NO WALL UNDER IT. It is carried as a LINE LOAD on the pressure slab, not as a support - 26 % (two-way) / 41 % (one-way bound). Robustness band: 4 No. T25 ADDITIONAL top and bottom over a 1200 width beneath HW3, lapped 1250 (50 phi) - mark S16, SHEET 06.
- ROOF SHEAR LINKS ARE MANDATORY AND WERE NOT IN THE PHASE 1 REPORT. tau_v 1.514 > tau_c 0.502, tau_c read at the STATIC M35 value because IS 4991 Cl. 10.3.1.1 forbids any dynamic increase on shear. T12 4-legged at 175 over the end 1200 each side, at 250 elsewhere.
- ONLY 300 mm OF WALL SITS ABOVE THE DOOR - too shallow for a lintel. Door head (-)2.000 + 2.100 = (+)0.100; roof soffit (+)0.400. The 300 wall and the 500 roof act TOGETHER as an 800 deep edge band; 4-T20 top + 4-T20 bottom, over the opening and 600 each side.
- THE DOOR FRAME IS CAST IN AND WELDED TO THE CAGE (EMP), even though the leaf is a security door and is not blast rated.
- HAUNCH 300 x 300 AT ALL FOUR WALL / ROOF JUNCTIONS, diagonal T16 @ 150. Top steel is continuous over every support, anchored Ld 800 into the wall. No opening occurs in the headhouse roof - the stair void is in the FLOOR, not the roof.

UNDERGROUND CBRN-HARDENED BLAST-RESISTANT PROTECTIVE STRUCTURE + SENTRY POST PUNE, MAHARASHTRA

NOTES

- ALL DIMENSIONS ARE IN MM. ALL LEVELS ARE IN METRES RELATIVE TO FINISHED SITE GRADE 0.000, NEGATIVE DOWNWARDS.
- WRITTEN DIM TO BE FOLLOWED ONLY. DRG NOT TO BE SCALED.
- THIS DRG IS TO BE READ IN CONJUNCTION WITH ALL RELEVANT ARCH AND STR DRGS.
- CONCRETE M35, w/c NOT MORE THAN 0.45. REINFORCEMENT Fe500D TO IS 1786:2008. Ld 40 PHI, LAPS 50 PHI, ALL STAGGERED SO THAT NOT MORE THAN 50 % ARE SPICED AT ONE SECTION.
- MAX BAR SPACING 150 IN BOTH CURTAINS - THIS IS AN EMP REQUIREMENT AND IS STRICTER THAN IS 456 Cl. 26.3.3.
- THE HEADHOUSE ROOF IS FLUSH WITH THE BERM CREST AT (+)0.900 AND CARRIES NO EARTH COVER. IT IS LOADED AT 396.5 kPa = 383 BLAST + 12.5 SELF + 1.0 SIDL.
- THE FOUR WALLS ARE REINFORCED SYMMETRICALLY FOR 383 kPa ACTING ON EITHER FACE - SEE THE CONSTRUCTION REQUIREMENTS PANEL. THIS IS A STATED REQUIREMENT, NOT AN ACCIDENT OF DETAILING.
- THE HEADHOUSE FLOOR IS THE TOP OF THE 900 PRESSURE SLAB AT (-)2.000. WALL STARTERS H10 ARE CAST WITH THAT SLAB - SEE SHEET 06.
- CONTRACTOR TO CHECK AND VERIFY THE DRG BEFORE EXECUTION.

STRUCTURAL REINFORCEMENT DETAILING - ENTRY HEADHOUSE ROOF & WALLS

CBRN HARDENED UG OPS ROOM

SHEET 14

Project number	01
Date	18 SEP 2026
Drawn by	SYN 01
Checked by	DR IR CHAUDHARI
STR014	
Scale	As indicated